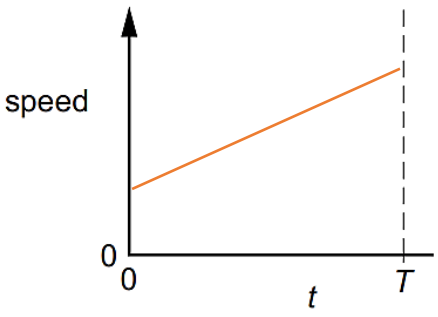


1	(a)	Using $v^2 = u^2 + 2as$ $8.7^2 = u^2 + 2(9.81)(1.5)$ $u = 6.8 \text{ m s}^{-1}$	C1 A1
	(b)	 Linear line and when $t = 0$, speed is more than 0	B1
	(c)	When the ball hits the ground, <u>it exerted a (normal) force on the ground.</u> By Newton's third law, <u>the ground exerts a (normal) force on the ball.</u> The <u>two forces are equal in magnitude but opposite in direction.</u>	B1 B1
	(d)(i)	Change in momentum = final momentum – initial momentum $= (0.059)(5.4) - (0.059)(-8.7)$ $= 0.83 \text{ N s}$	C1 A1
	(d)(ii)	Resultant force on the ball $= 0.83/0.091$ $= 9.1 \text{ N}$ Force by ground on ball $= 9.1 + (0.059)(9.81)$ $= 9.7 \text{ N}$	C1 C1 A1
	(e)	The <u>resultant force will be equal to weight – air resistance. Air resistance increases with speed.</u> Therefore, the resultant force and acceleration will decrease with time. Thus, the <u>gradient of the graph will decrease with time.</u>	B1 B1

2	(a)	For an object to move in circular motion at constant speed, it only changes	
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